

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills,Kumaraswamy Layout,Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“Full Stack Web Development Intern”

Submitted in partial fulfillment for the award of degree (20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submittedby:

Samit Chittiappa Malachira-1DS20AI051

Conducted at



Skolar

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**FULL-STACK WEB DEVELOPMENT INTERN**” carried out by **Mr Samit Chittiappa Malachira[1DS20AI051]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

**Signature of
Reporting Head**
Mohammad Toib
Skoalr

**Signature of
Internship
Coordinator**
Prof. Kavya D N
Dept. of AI&ML
DSCE

Signature of HoD
Dr.Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

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Department of Artificial Intelligence and Machine Learning**



D E C L A R A T I O N

I, **Samit Chittiappa Malachira**, final year student of Artificial Intelligence And Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in Skolar. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date:

USN:1DS20AI051

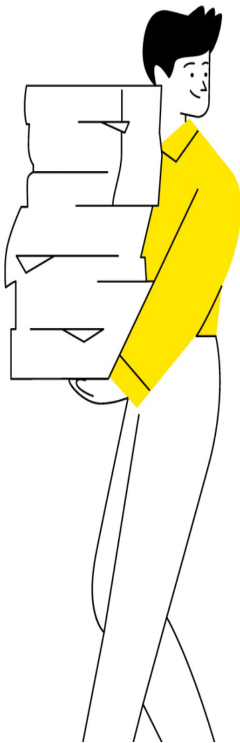
Place:Bangalore

NAME:Samit Chittiappa Malachira

OFFER LETTER

27/01/2023

Welcome Learner!



Dear **samit chittiappa malachira**,

This is to certify that you have been enrolled for
Full stack web development internship program
at Skolar starting from **February**.

We hope this course becomes your ticket to
stations with countless opportunities.

A stylized, handwritten signature in black ink, appearing to read 'Taufique Ahmad'.

Taufique Ahmad
Founder & COO

COMPLETION CERTIFICATE



Internship Completion Certificate

This is to certify that

samit chittiappa malachira

From **dayananda sagar college of engineering** has successfully completed a **3 months** internship and training on **Full Stack Web Development** from **01 February 2023** to **30 April 2023**

samit chittiappa malachira has shown exceptional results in assessments and evaluations.

We wish him/her all the best for the future.

Muskan Rastogi
Founder and CEO

Date : 02/05/2023
Certificate number: 22-1393

Mohammad Toib
Course coordinator

AC K N OWLED G EMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Mohammad Toib, Skolar, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor–AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Samit Chittiappa
Malachira[1DS20AI051]

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CHAPTER 1

INTRODUCTION

During my recent full-stack web development internship, I embarked on an enriching journey that encompassed a comprehensive array of topics, ranging from foundational concepts to advanced methodologies. Over the course of the program, I immersed myself in a structured curriculum designed to provide a holistic understanding of full-stack development and its practical applications.

The internship commenced with a thorough exploration of basic concepts in web development, laying the groundwork for more advanced topics. I delved into the fundamentals of front-end development, grasping the nuances of building user interfaces and understanding the underlying principles of HTML, CSS, and JavaScript. This served as a precursor to the study of back-end development, where I learned to design and implement server-side logic and database interactions, employing technologies such as Node.js, Express.js, and MongoDB.

As the internship progressed, I delved deeper into the realm of front-end frameworks and libraries, exploring technologies like React.js and Vue.js. Through hands-on exercises and guided projects, I honed my skills in building interactive and responsive web applications, gaining proficiency in state management and component-based architecture.

The exploration of back-end development opened new horizons, allowing me to design and implement robust APIs and microservices. I delved into topics such as RESTful architecture and authentication mechanisms, mastering the art of building scalable and secure server-side applications.

The internship also provided an opportunity to delve into the fascinating world of full-stack frameworks and platforms. I gained insights into the architecture and functioning of frameworks like Django and Ruby on Rails, exploring concepts such as MVC (Model-View-Controller) and ORM (Object-Relational Mapping). Through practical exercises, I honed my skills in building and deploying full-stack applications, unlocking their potential for complex web development tasks.

During my recent full-stack web development internship, I had the opportunity to collaborate with a team of five members on an exciting project focused on developing a restaurant ordering website. In this project, my role primarily revolved around database management using SQL and assisting in the design of the website's user interface.

In summary, my full-stack web development internship provided a comprehensive learning experience, equipping me with the knowledge, skills, and practical experience to tackle diverse challenges within the realm of web development. From mastering fundamental concepts to delving into advanced frameworks and platforms, and applying them to real-world projects, the internship served as a transformative journey that deepened my understanding and passion for full-stack development.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Skolar is a pioneering E-learning platform in India, revolutionizing education by emphasizing practical, outcome-driven learning. Since its creation in March 2020, Skolar has been committed to bridging the gap between traditional education and industry demands. Specializing in domains such as Computer Science, Electronics and Communications, Mechanical/Civil Engineering, Management, and Biotechnology, Skolar offers comprehensive programs tailored to meet the evolving needs of learners.

With a focus on empowering students to upskill and excel in their careers, Skolar provides high-quality mentorship and guidance, ensuring individuals have the practical knowledge and skills necessary to thrive in today's competitive job market. Through leveraging cutting-edge technology, Skolar delivers dynamic and immersive learning experiences, enabling learners to gain hands-on industry experience from anywhere, anytime. Having positively impacted over 1 lakh students, Skolar is dedicated to transforming the learning landscape by providing a platform where individuals can unleash their full potential and succeed in the rapidly evolving world of work.

CHAPTER 3

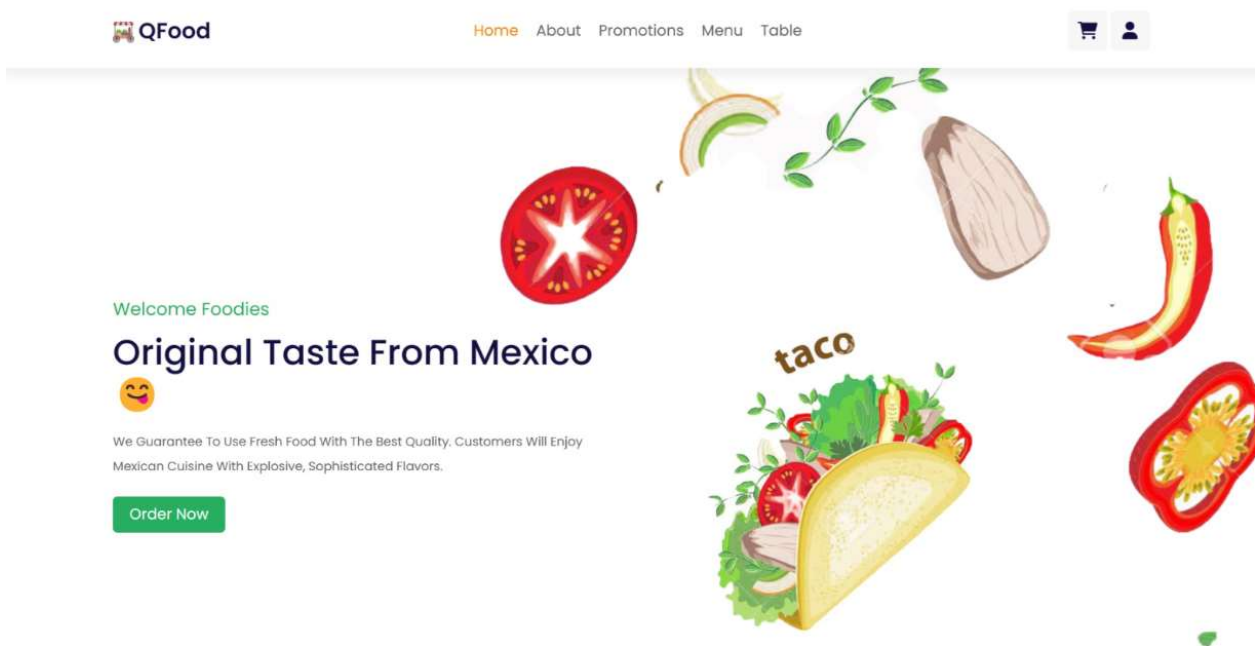
SCOPE OF WORK

- 1. Project Management:** As the project manager, responsibilities include overseeing the project timeline, ensuring timely delivery of milestones, and effectively allocating resources among team members. Coordination of tasks and communication among team members is crucial to keep everyone aligned with project goals. Additionally, addressing any issues or roadblocks promptly and ensuring adherence to project requirements and quality standards are essential for the smooth progress of the project.
- 2. Front-end Development:** In front-end development, the focus lies in crafting the user interface of the website using HTML, CSS, and JavaScript. This involves translating design mockups and wireframes into interactive web elements that provide an optimal user experience. Collaborating closely with UI/UX designers, the aim is to ensure cross-browser compatibility and adherence to accessibility standards, allowing users to seamlessly interact with the website across different platforms and devices.
- 3. Back-end Development:** Back-end development entails building the server-side logic and database interactions that power the website's functionality. Leveraging technologies like Node.js, Express.js, and SQL, the goal is to handle user requests efficiently and manage data seamlessly. This involves developing robust APIs and endpoints for various functionalities, such as user authentication, order processing, and data retrieval. Ensuring data integrity and optimizing database performance are critical aspects of back-end development.
- 4. Database Management:** Database management involves designing and maintaining the database architecture to efficiently store and organize data. This includes creating database schemas, establishing relationships between tables, and optimizing queries for

performance. With a focus on data integrity and scalability, database management aims to ensure that the database can handle the website's data requirements effectively and efficiently.

5. Order Management: Order management encompasses the development of features related to order processing, confirmation, and real-time tracking. This includes functionalities such as allowing users to place orders, receive order confirmations, and track order status. Integrating payment gateways for secure transactions is also a key aspect of order management, ensuring a seamless and secure checkout process for users.

6. Search and Filtering: Enhancing user experience through robust search functionality and intuitive filtering options is the focus of search and filtering. This involves implementing search algorithms that accurately retrieve relevant results based on user queries and developing filtering options based on various criteria such as category, price range, or dietary preferences. The goal is to empower users to easily find the information they're looking for and make informed decisions while browsing the website.



CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

1. Front-end Development: Gain practical experience in applying front-end development technologies such as HTML, CSS, and JavaScript to create responsive and visually appealing user interfaces.

2. Back-end Development: Develop proficiency in back-end development utilizing frameworks like Node.js and Express.js to build robust server-side logic and RESTful APIs for efficient data processing and management.

3. Database Management: Acquire skills in database management, including designing schemas, writing optimized SQL queries, and ensuring data integrity, to support scalable and reliable web applications.

4. Project Management: Learn about project management practices, including timeline management, resource allocation, and issue resolution, to effectively coordinate team efforts and ensure project success.

5. User Experience Design: Understand the importance of user experience design principles and accessibility standards in creating intuitive and inclusive web interfaces for diverse user demographics.

6. Integration of Third-party Services: Explore the integration of third-party services such as payment gateways and APIs to enhance the functionality and security of web applications, while adhering to industry best practices.

7. Agile Development Methodologies: Gain insights into agile development methodologies and collaborative workflows, including version control systems like Git, to streamline development processes and foster team collaboration.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Challenges:

- 1. Cross-browser Compatibility:** Ensuring that the website functions correctly across different web browsers (e.g., Chrome, Firefox, Safari, Internet Explorer) can be challenging due to variations in rendering engines and standards compliance.
- 2. Scalability and Performance:** As the website's user base grows, ensuring scalability and optimal performance becomes crucial. Heavy traffic or inefficient database queries can lead to slow loading times and degraded user experience.
- 3. Data Security:** Protecting sensitive user information, such as login credentials and payment details, from security breaches and unauthorized access is paramount.
- 4. User Experience Optimization:** Designing an intuitive and user-friendly interface that meets the needs of diverse user demographics can be challenging.

Solutions:

- 1. Cross-browser Compatibility:** To address cross-browser compatibility challenges, it's important to regularly test the website across multiple browsers and utilize browser testing tools. Following standards-compliant coding practices is also essential to ensure consistency in rendering across different browsers.
- 2. Scalability and Performance:** Tackling scalability and performance issues involves optimizing database queries, implementing caching mechanisms, employing load balancing, and monitoring server performance regularly. This ensures that the website can handle increasing traffic efficiently without compromising on performance.

- 3. Data Security:** To enhance data security, implementing encryption protocols, secure authentication mechanisms, input validation, and conducting regular security audits are key measures. These steps safeguard sensitive user information from unauthorized access and breaches.
- 4. User Experience Optimization:** For user experience optimization, conducting user research to understand user preferences, utilizing responsive design principles for adaptability across devices, following accessibility standards for inclusivity, and implementing user-friendly features can significantly enhance the overall user experience.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Individual Achievements:

- **Mastery of Full-Stack Technologies:** During the internship, mastery of various full-stack web development technologies and tools was achieved, including front-end frameworks like React.js or Angular, back-end frameworks like Node.js or Django, database management systems, and version control systems.
- **Successful Project Delivery as Part of a Team:** Contributed to a significant project as part of a team, resulting in the successful development and deployment of a web application. Met project milestones and deadlines while delivering high-quality results.
- **Certifications Obtained:** Obtained certifications relevant to full-stack web development, such as Full-Stack Web Developer Certification, or specialized certifications in front-end or back-end technologies.

Contributions to Team/Organization:

- **Enhancements to Web Development Processes:** Leveraged insights gained during the internship to enhance the organization's web development processes. Implemented best practices, optimized workflows, and introduced automation tools to improve efficiency and productivity.
- **Collaborative Project Contribution:** Played a key role in a collaborative web development project, contributing to its successful outcome through effective teamwork, communication, and problem-solving. Demonstrated leadership and technical expertise to drive project success and achieve organizational goals.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my internship experience in full-stack web development and the restaurant online ordering system project, I'm grateful for the opportunity to have contributed to the project, primarily focusing on the database aspect using MySQL and assisting in designing the website UI as part of a collaborative team effort.

During the internship, I had the chance to deepen my understanding of database management by working extensively with MySQL. From designing efficient database schemas to implementing complex queries to optimize data retrieval, this experience provided me with invaluable hands-on experience in managing the data backbone of web applications.

Additionally, I played a role in shaping the user interface of the restaurant online ordering system. Collaborating closely with my team members, I contributed to the design process, offering insights and suggestions to create a user-friendly and visually appealing interface that enhances the overall customer experience.

Though my contributions were primarily focused on the database and UI design aspects, I also had the opportunity to learn from other team members and gain exposure to different components of full-stack web development, further enriching my internship experience.

Of course, the journey presented its share of challenges. From ensuring data integrity and optimizing database performance to refining UI elements for seamless user interaction, each obstacle served as a learning opportunity, fostering my growth as a developer.

Looking ahead, I'm excited to continue honing my skills in full-stack web development, building upon the foundation laid during the internship. I'm eager to explore new technologies and methodologies, expand my expertise across various aspects of web development, and contribute to impactful projects that leverage technology to solve real-world problems.

In conclusion, my internship experience in full-stack web development and the restaurant online ordering system project provided me with valuable insights and hands-on experience, particularly in database management and UI design. I'm grateful for the opportunity to have contributed to the project and look forward to applying my skills and knowledge to future endeavors in the field.

CHAPTER 8

RECOMMENDATIONS

In conclusion, this virtual internship has significantly enriched my understanding and proficiency in full-stack web development. The hands-on tasks and challenges presented throughout the program have been instrumental in bridging the gap between theoretical concepts and their practical implementation. Engaging directly with real-world scenarios has provided me with a solid foundation in full-stack development, enabling me to effectively navigate and solve complex problems.

The mentorship aspect of the internship has been particularly valuable. The mentors offered personalized feedback and guidance that proved crucial to my learning journey. Their expertise in full-stack development not only deepened my understanding but also bolstered my confidence as a developer. Through constructive critiques and one-on-one sessions, I was able to refine my techniques and strategies in tackling various web development projects.

Overall, this internship offers a comprehensive and immersive learning experience that is both challenging and rewarding. It serves as an excellent opportunity for individuals aspiring to become proficient full-stack developers. The structured yet flexible curriculum caters to individuals at various skill levels, ensuring that all participants gain the necessary expertise to excel in the field of full-stack web development. I highly recommend this program to anyone seeking professional growth and expertise in this dynamic field.

CHAPTER 9

CONCLUSION

The full-stack web development internship has been a transformative journey, significantly enhancing my proficiency in creating dynamic and engaging online platforms. Throughout the program, I delved into a comprehensive curriculum covering both the front-end and back-end aspects of web development, gaining a deep understanding of HTML, CSS, JavaScript, server-side scripting, and database management. The hands-on projects, notably the creation of a restaurant online website, provided invaluable practical experience, allowing me to apply theoretical knowledge to real-world scenarios.

One of the most rewarding aspects of the internship was working with a team of five on the restaurant online website project. Collaborating closely with my team members, we leveraged a variety of technologies to develop a fully functional and visually appealing platform that catered to both customers and restaurant owners. From designing intuitive user interfaces to implementing secure payment gateways and managing backend operations, every aspect of the project contributed to our growth as full-stack developers.

Working within a team environment not only enhanced the learning experience but also provided valuable insights into effective teamwork, communication, and project management. By leveraging each other's strengths and expertise, we were able to overcome challenges and deliver a high-quality product within the allotted timeframe.

As I conclude this internship, I am grateful for the opportunities it has provided and the skills I have acquired. I am excited to apply my newfound knowledge and expertise to future web development projects, confident in my ability to thrive both independently and as part of a collaborative team in the dynamic world of full-stack web development.

CHAPTER 10

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